

P4B Solar Mini-Grid: Socioeconomic Impact Analysis

CS 6150 – Computing for Good
Fall 2024

Members and Roles

- Sophia Wu
 - Project Manager, Lead Front-end Developer, Full-stack Developer
- Brock Pace
 - Lead Back-end Developer, Full-stack Developer
- Bokyoung Won
 - Data Analyst and Scientist
- Xiaobo Ren
 - Data Analyst and Scientist

Solar Mini-Grid: Socioeconomic Impact Analysis

This project aims to draw new insights and conclusions through further analysis of the data provided by our partner by focusing on the subgroups of education and sharing the findings with stakeholders.

Planned Deliverables:

- Website containing visualizations and infographics with analyses and correlations to relate mini-solar-grids and their impact on the surrounding community.
- A subgroup analysis framework, methodology outlining the criteria for analyzing specific subgroups (e.g., education), including demographic filters, statistical approaches, hypotheses and analytic approaches.
- Analysis report to include the preparation, implementation, validation and conclusion of the analysis, the insights of the social impact based on the subgroup and what further analysis is needed.
- Cleaned dataset generated from the raw and any other external data we leveraged in this analysis, this is the basement of the analysis we will implement.

Partner Organization

RENEWVIA

SOLAR AFRICA

www.renewvia.com

Nicholas Selby

VP of Engineering

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Timeline with Mileposts

Date	Milepost
09/29/2024	P4 Mini Presentation
10/04/2024	Data Analysis Framework
10/11/2024	Cleaned Dataset prepared for analysis
10/30/2024	Initial Data Analysis Completed
11/05/2024	Key Result Finalized
11/10/2024	Initial Analysis Report include analysis implementation and result
11/17/2024	Website Completed
11/24/2024	Final Presentation
12/01/2024	Developer Documentation
12/08/2024	Final Analysis Report
12/15/2024	Final Presentation

Architecture Details

- Website Front-end

- Vercel: Platform that allows building, deployment, and hosting of websites and web applications
- Next.js: React framework with ready-made features and components that allows developers to quickly develop websites and web applications
- Typescript: Programming language popular for building large or complex websites and apps

- Back-end API

- Java: Spring Boot

- Data Analysis

- Python : Organizing and handling large dataset, visualizing data trends, mathematical operations, and performing machine learning modes.
- Java: Spring Data, SQL
- Tableau: data visualization (Creating dashboards)

Project Home Page

<https://cs6150-solar-grid-socioeconomic-impact.vercel.app/>

CS6150 Computing for Good

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Team P4B

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Project Description

Earlier this year in 2024, Renewvia published the first-ever peer-reviewed, comprehensive socioeconomic impact analysis for mini-grids in rural Africa. The data is open-source, and the statistical tools used in the paper to draw conclusions are all very general. Little attention was paid to subgroup analyses or correlations in results. There is great potential for follow-up analyses to ask and answer interesting questions related to the impact of Renewvia's work on communities and how better understanding that impact could influence the engineering design process.

Project Goal

Our group aims to draw new insights and conclusions through further analysis of the data provided by Renewvia by analyzing subgroups and observing possible correlations in the results of the review. These findings will be displayed through data visualizations and written observations on the website. From there, we aim to observe the impact of solar mini-grids on the surrounding community based on our analyses.

Team Members and Planned Roles

Sophia Wu - Project Manager, Front-end Developer, Full-stack Developer

Brock Pace - Back-end Developer, Full-stack Developer

Bokyoung Won - Data Analyst and Scientist

Xiaobo Ren - Data Analyst and Scientist

Samuel Walden - Back-end Developer, Data Analyst

Lighthouse Scores

Desktop

Performance: 100

Accessibility: 100

Best Practices: 100

SEO: 100

Mobile

Performance: 99

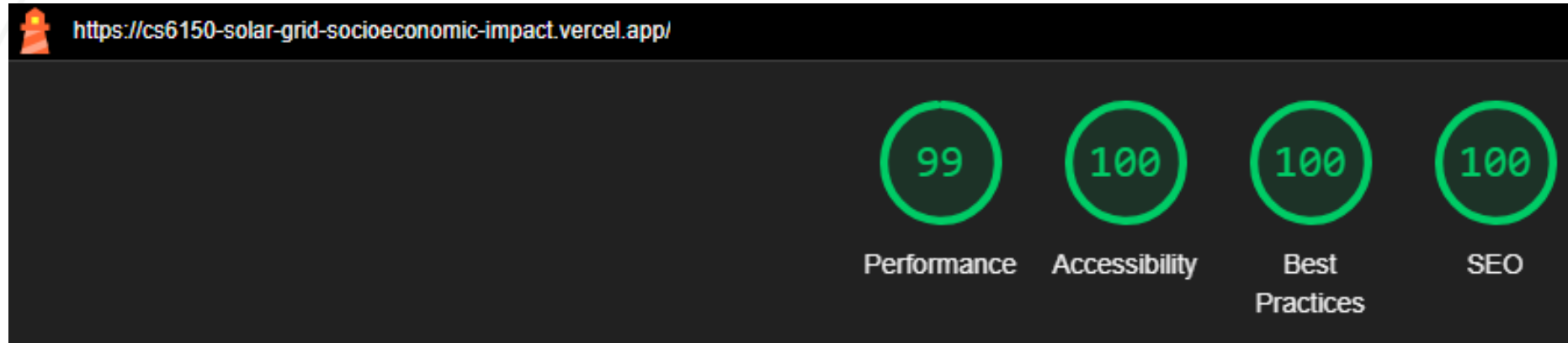
Accessibility: 100

Best Practices: 100

SEO: 100

Project Website – Lighthouse Metrics

- Mobile



- Desktop

